

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

03rd November 2014

**Subject ST4 - Pensions and Other Employee
Benefits**

Time allowed: Three hours (14.45* – 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
3. *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
4. *The answers are not expected to be any country or jurisdiction specific. However, if Examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
5. *Attempt all questions, beginning your answer to each question on a separate sheet.*
6. *Mark allocations are shown in brackets.*
7. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

Q. 1) A new Government of a Country has been elected with a large majority and the population has great expectations from the Government. As a part of their manifesto, the Government plans to introduce social security benefits for retirement.

- i)** List the roles the Government play in relation to benefit provision. (3)

You are an actuary advising the Government on the benefit provision.

You have suggested that the Government scheme should be used to target benefits to those most in need and that the scheme should provide little or no benefit to individuals with private means. In this way, resources will be directed to the less well off or those deemed to be in need.

- ii)** Set out the advantage and disadvantages of this approach. (5)

The Government plans to launch a National Pension Scheme (NPS) where individuals contribute to the scheme and the Government also contributes an equal amount for the first five years to those who contribute up to Rs. 1000. The contributions can be invested in any fund chosen by the individuals based on their circumstances. Each fund option will have minimal fund management charges. The individual's accumulated fund can be used to purchase an annuity from an insurance company at retirement.

- iii)** Suggest reasons why the Government may have chosen this NPS and list its disadvantages. (7)

- iv)** List the factors that affect the risk profile of any benefit scheme. (2)
[17]

Q. 2) The Government of a developing country is considering setting standards for the disclosure of information regarding pension liabilities in the sponsoring company's accounts.

- i)** Suggest reasons why such standards may be desirable. (3)

- ii)** List the main items which might be disclosed in respect of a final salary pension scheme. (4)
[7]

Q. 3) The trustees of a defined benefit pension scheme are planning to assess the strength of the sponsoring employer's covenant. One of the trustees is also the Finance Director and the other trustees have asked him to assess the employer's covenant and to undertake the ongoing monitoring process on behalf of the trustees.

- i)** Outline why the assessment of the employer covenant is important. (2)

- ii)** Outline the key issues the Finance Director should consider in undertaking this task. (4)

- iii)** List the information that could be used to assess the employer's covenant. (2)

The current scheme is under-funded and the trustees have to decide on the employer contribution to be paid in the next five years.

- iv) Discuss the issues the trustees will consider when deciding on the level of employer contribution to be paid.

(6)
[14]

- Q. 4)** As an alternative to direct investment in assets, it may be possible to obtain an investment product in which the product provider determines the choice of assets. Insurance companies, banks or other financial institutions may provide these products.

Set out three examples of these types of products and describe their key characteristics.

[6]

- Q. 5)** i) Write down the mathematical expression for all four methods of funding for Actuarial Liability (AL) and Standard Contribution Rate (SCR).

(2)

- ii) Sketch a graph showing the development of the SCRs over time, for each funding method, for a scheme that starts with all the members at the assumed entry age (and no past service), then eventually stabilizes in terms of age/sex/salary and past service structure.

(4)
[6]

- Q. 6)** You are the consulting actuary of a large defined benefit pension scheme and are about to undertake a formal actuarial funding valuation. You have been advising the trustees on the assumptions for the funding valuation.

The scheme trustees are specifically interested in understanding how the discount rates used to value the assets and liabilities and mortality assumptions are set.

The liabilities of the scheme are approximately Rs 4 crore for pensions in payment and Rs 6 crore for non-pensioners. The scheme is currently invested 80% in equities and 20% in government bonds.

The following current market information and historic data is available:

Gross Redemption yield on over 15-year government bonds: 8.7% per annum
Gross Redemption yield on under 15-year government bonds: 8.2% per annum

Period of measurement	Annual average return, in excess of inflation	
	on Equities	on Bonds
1964 to 2013	6.00%	0.90%

Mortality table used for previous valuation – PMA92

- i) For each of the three approaches to setting the discount rate listed below explain the key features of the approach and suggest an appropriate discount rate, based on the information provided:

- Asset-based discount rate
- Mark to market
- Bond yield plus risk premium

(12)

- ii) Discuss the various methods in which the mortality tables can be projected into future.

(6)

The scheme sponsor wishes to reduce investment risk when market conditions allow. The scheme is proposing to put in place a strategy to reduce risk over the longer term by establishing a process whereby the percentage equity holding is automatically reduced in stages when pre-arranged funding levels are reached.

iii) Discuss the advantages and disadvantages of this proposed strategy.

(5)

[23]

Q. 7) Mr Alpha, a 25 year old employee of Infinity Limited is a member of the company sponsored Defined Benefit (DB) scheme with an accrual rate of 2% and normal retirement age of 65. Infinity Limited has decided to close the DB scheme and set up a money purchase (MP) pension scheme.

Infinity Limited has decided that they will make the following age related contributions to the new MP scheme:

Age Last Birthday	Employer contribution (% of basic salary)
Under 30	5
30 to 39	7
40 to 49	9
50 and over	12

Ignoring pre-retirement decrements and using the following assumptions:

Salary increases = price inflation + 1.75% per annum

Interest = price inflation + 5% per annum

Annuity cost at age 65 = 15 for each unit of pension

i) Calculate how much Mr. Alpha should contribute to the MP scheme so as to get the same benefit as under the previous DB scheme assuming he stays in service until age 65.

(7)

Mr Alpha has raised a concern that his contribution will increase significantly if he wishes to maintain the same level of benefit at retirement as the DB scheme. Infinity Limited is considering whether they should offer some guarantees to the members of the new MP scheme.

ii) Briefly describe the form of guarantees that could be provided by Infinity Limited.

(2)

[9]

Q. 8) A developing country has Defined Contribution (DC) pension provision for its residents. The residents are allowed to retain some (or all) of their DC funds in their own personal investment account. After retirement residents are able to withdraw money from these accounts to meet ongoing needs.

i) Explain the risk that a resident faces if he keeps the money invested in assets of his own choosing and draws an income from the fund every year.

(3)

The government is concerned that the residents will run out of funds for retirement by not purchasing an annuity with an insurer at age 65. The government imposes a requirement for its residents to use the whole of their DC funds to purchase an annuity with an insurer at age 65. This policy is unpopular with residents of the country because the annuities are viewed as expensive and inflexible.

- ii) Discuss the options that the government may consider in order to address residents' concerns.

(7)
[10]

Q. 9) An employer offers a generous life assurance scheme to its staff. The nature of the employment is such that mortality rates are higher than average. The benefit paid is a lump sum of 10 times salary. The cost of the scheme is a material part of the employer's total operating costs.

- i) Suggest alternative methods of financing the life assurance benefit.
- ii) Explain how the annual cost to the employer would be calculated in each case described above.

(3)

(5)
[8]
